Selective Local Filtering and Finite-Cycle Algebra:
Exact Selection Boundaries in a Conditional Quantum-Composition Family

Kirtus Justus
Research direction and responsible authorship
Port Coquitlam, British Columbia, Canada
AI-assisted internal mathematical analysis: ChatGPT (GPT-6, OpenAI)

12 November 2026, B.C. Pacific Time
Initial External Grounding open; no total ontology selected

Abstract

A previously constructed convex two-system family agrees with standard elementary complex quantum geometry while leaving composite preparations undetermined. We show that every strict intermediate member is additionally *not closed under selective local filtering*: a two-site product filter, with strictly positive success probability, produces a state beyond the family's exact Bell-fidelity bound. The witness is reconstructible from single-site measurements, requiring no entangling measurement as an experimental premise. Under a further topological closure assumption, allowing all local filters and local unitaries forces the entire conventional two-system density-matrix set, making the operational cost of this selection explicit. Separately, the imaginary-algebra seed from a directed three-cycle extends to every odd directed cycle at the level of a nondegenerate alternating contrast form, after a stated linearization; even cycles provide a null-mode control. Neither construction derives complex amplitudes, quantum instruments, Born probability, physical dynamics, classical mechanics, nor an empirical prephysical-to-physical map. All quantum notation below is a *stipulated mathematical test framework*, not an identified physical qubit.

1 Defined scope and a source-controlled adversarial question

Let $H = \mathbb{C}^2$ be an *assumed* complex two-dimensional Hilbert space. This mathematical system is not an operationally identified two-level system or a derived physical qubit. Let $\mathcal{D}_2$ be the normalized positive semidefinite operators on $H \otimes H$, and Sep_2 its separable convex subset. For $t \in [0,1]$, define the conditional toy family

$$K_t = (1-t)\mathrm{Sep}_2 + t\mathcal{D}_2 = \{(1-t)\sigma + t\tau : \sigma \in \mathrm{Sep}_2, \tau \in \mathcal{D}_2\}. \tag{1}$$

These compact convex sets contain all product states and are invariant under local *trace-preserving* quantum channels, but they are not asserted to admit every *outcome-conditioned* quantum operation. The question tested here is the consequence of *adding* selective single-site filters, not whether they were logically entailed by K_t or by a prephysical primitive. The standard trace probability is also imported, not derived.
Set $|\Psi\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$, $P_\Psi = |\Psi\rangle\langle\Psi|$ and $P_{00} = |00\rangle\langle 00|$. The state

$$\rho_t = (1-t)P_{00} + tP_\Psi \in K_t \tag{2}$$

is an allowed preparation at every t. For any *normalized* product vector $|a\rangle \otimes |b\rangle$, Cauchy–Schwarz gives $\langle ab|P_\Psi|ab\rangle \le \frac{1}{2}$. Convexity and $\mathrm{Tr}(P_\Psi \tau) \le 1$ therefore imply

$$\sup_{\rho \in K_t} \mathrm{Tr}(P_\Psi \rho) = \frac{1+t}{2}, \tag{3}$$

where the upper bound is attained by $(1-t)|01\rangle\langle 01| + tP_\Psi$; the starting state (2) itself need not attain it. This exact supporting-functional inequality is the discrimination criterion.

2 A finite-probability selective-instrument counterexample

Theorem 1 (Failure of selective local-filter closure)**.** *For every $0 < t < 1$, there exists a product of two physically conventional single-site filter branches whose normalized successful output from an allowed $\rho_t \in K_t$ lies outside K_t. One may choose $F_\varepsilon = \mathrm{diag}(\varepsilon, 1)$ on both sites, with $0 < \varepsilon^2 < t/(1+t)$.*

Proof. Put $x = \varepsilon^2$. The successful branch has Kraus operator $F_\varepsilon \otimes F_\varepsilon$. It maps $|00\rangle$ to $\varepsilon^2|00\rangle$ and $|\Psi\rangle$ to $\varepsilon|\Psi\rangle$. Hence the positive success probability and the normalized output are

$$p_{\mathrm{succ}} = (1-t)x^2 + tx = x[t + (1-t)x] > 0, \tag{4}$$

$$\rho'_{t,x} = \frac{(1-t)xP_{00} + tP_\Psi}{t + (1-t)x}, \qquad \mathrm{Tr}(P_\Psi \rho'_{t,x}) = \frac{t}{t + (1-t)x}. \tag{5}$$

Subtracting the sharp upper bound (3) and factoring gives

$$\delta(t,x) = \frac{(1-t)[t - (1+t)x]}{2[t + (1-t)x]} > 0 \quad \Longleftrightarrow \quad 0 < x < \frac{t}{1+t} \tag{6}$$

for each strict intermediate t. Thus the conditional state is not in K_t. The success branch is a legitimate *product* operation: F_ε can be complemented by $G_\varepsilon = \mathrm{diag}(\sqrt{1-\varepsilon^2}, 0)$, with $F_\varepsilon^\dagger F_\varepsilon + G_\varepsilon^\dagger G_\varepsilon = \mathbf{1}$. Sequential local filtering supplies the successful joint branch without an entangling interaction. □

Exact finite counterexample. At $t = \frac{1}{2}$ and $\varepsilon = \frac{1}{2}$, success occurs with probability $5/32$ and the conditioned state is $\rho' = (P_{00} + 4P_\Psi)/5$. It has $\mathrm{Tr}(P_\Psi \rho') = 4/5$, whereas the maximum inside $K_{1/2}$ is $3/4$, a *strict rational discrepancy of* $1/20$. No asymptotic limit or numerical rounding is needed.

Proposition 2 (The discrepancy is detectable with elementary product measurements)**.** *Writing X, Y, Z for conventional single-site Pauli observables, the separating projector obeys*

$$P_\Psi = \tfrac{1}{4}(\mathbf{1} \otimes \mathbf{1} + X \otimes X + Y \otimes Y - Z \otimes Z). \tag{7}$$

For the exact example, the three correlations are $(\langle XX\rangle, \langle YY\rangle, \langle ZZ\rangle) = (4/5, 4/5, -3/5)$. Consequently the violation of the $K_{1/2}$ bound is reconstructible from three settings of local *measurements.*

Proof. Equation (7) follows by direct multiplication in the computational basis. Inserting the two orthogonal projectors P_{00} and P_Ψ gives the stated correlations and the fidelity $\frac{1}{4}(1 + \frac{4}{5} + \frac{4}{5} + \frac{3}{5}) = \frac{4}{5}$. The state remains outside the set whether or not a Bell-projector joint measurement is included among its effects. □

A quantitative cost, not an empirical frequency. For a uniform exact choice $x = t/[2(1+t)]$, the positive discrepancy and success probability are

$$\delta = \frac{1-t^2}{2(3+t)}, \qquad p_{\mathrm{succ}} = \frac{t^2(3+t)}{4(1+t)^2}. \tag{8}$$

Within *this one-parameter symmetric-filter family*, strict escape requires $x < t/(1+t)$ and therefore $p_{\mathrm{succ}} < 2t^2/(1+t)^2$. Thus the successful branch can be rare when t is small; this is not a bound on all imaginable filters. At $t = 0$, the chosen uniform filter has zero success and is *undefined* as a conditional state; at $t = 1$, $K_t = \mathcal{D}_2$ and no violation occurs. Both endpoints are retained as controls.

3 What the added instrument premise actually selects

Theorem 3 (Conditional two-system saturation)**.** *Fix $t \in (0, 1]$. Suppose $S \subseteq \mathcal{D}_2$ is a* closed, convex *set containing the state ρ_t of (2), invariant under all local unitaries and under* normalization of every nonzero-success product of local quantum filter branches. *Then $S = \mathcal{D}_2$.*

Proof. Applying the filters of Theorem 1 with $\varepsilon > 0$ gives $\rho'_{t,\varepsilon^2} \in S$. As $\varepsilon \downarrow 0$, this converges in trace norm to P_Ψ because $t > 0$. Closedness of S puts P_Ψ in S. A local bit-flip on the second factor maps it to P_Φ, the projector on $(|00\rangle + |11\rangle)/\sqrt{2}$. For any $s_0, s_1 \geq 0$ with $s_0^2 + s_1^2 = 1$, the nonzero-success local filter $\mathrm{diag}(s_0, s_1) \otimes \mathbf{1}$ applied to P_Φ produces the pure Schmidt state $s_0|00\rangle + s_1|11\rangle$ after normalization. Arbitrary local unitaries and the Schmidt decomposition then supply every two-system pure-state projector. By convexity, their mixtures exhaust $\mathcal{D}_2$. The reverse inclusion was assumed; equality follows. □

The theorem is a logical *selection test* conditional on the pre-existing complex operator representation and complete local filtering/normalization, unboundedly low success probabilities, topological closure, local unitaries and mixing. It does not produce those structures from a finite relation. It also does not disprove an intermediate model that allows only deterministic local channels or restricts selective instruments. Local filtering and its entanglement-concentration consequences have a substantial pre-existing literature, including Verstraete, Dehaene and De Moor [1]; no priority claim is made for that general mechanism.

4 Independent control: odd finite cycles do not uniquely select quantum states

A separate algebraic check tests dependence on the *specific* three-cycle input. Let P_n cyclically permute the vertices of a timeless directed n-cycle; after the explicitly chosen rational linearization, put

$$V_n = \{v \in \mathbb{Q}^n : \sum_j v_j = 0\}, \quad C_n = P_n - P_n^{\mathsf{T}}, \quad \Omega_n(v, w) = v^{\mathsf{T}} C_n w.$$

Proposition 4 (Odd-cycle contrast dichotomy)**.** *For $n \geq 3$, Ω_n restricted to V_n is nondegenerate iff n is odd. Its skew-operator rank on $\mathbb{Q}^n$ is $n-1$ for odd n and $n-2$ for even n. For odd n, after choosing the real completion and the counting inner product, $J_n = C_n(-C_n^2)^{-1/2}$, defined on $V_n \otimes_{\mathbb{Q}} \mathbb{R}$, obeys $J_n^2 = -\mathbf{1}$ and $J_n^{\mathsf{T}} = -J_n$. Reversing the directed cycle sends J_n to $-J_n$.*

Proof. Over $\mathbb{C}$, the Fourier basis of P_n has eigenvalues $e^{2\pi ik/n}$ for $k = 0, \ldots, n-1$, so C_n has eigenvalues $2i\sin(2\pi k/n)$. The constant eigenvector ($k = 0$) is removed by V_n. A second zero mode $k = n/2$ exists iff n is even. This proves the rank and alternating-form assertions, since C_n is skew-symmetric. For odd n, $-C_n^2$ is positive definite on the real contrast space, whose positive spectral square root therefore exists. Functional calculus gives $J_n^2 = -\mathbf{1}$, $J_n^{\mathsf{T}} = -J_n$. A reflection conjugates P_n to P_n^{-1} and therefore C_n to $-C_n$ while preserving C_n^2, proving the orientation reversal. □

For $n = 3$ the stronger exceptional identity $C_3^2 = -3\mathbf{1}$ holds on the contrast plane; for odd $n \geq 5$, different nonzero Fourier magnitudes prevent C_n^2 from being a scalar multiple of $\mathbf{1}$. Thus the three-cycle's imaginary quadratic field is a special *algebraic simplification*, not evidence of physical selection. Every initial finite label set and its cycle action also supports a conventional classical simplex Δ_{n-1} with affine effects. The same algebraic contrast modes therefore underdetermine the operational completion. No cyclic edge is interpreted as a physical clock, oscillation, wave or frequency.

5 Scientific boundary and reproducibility

In the source research, the conditional holonomy determinant law presupposes quantum projective geometry. The filtering result concerns the independent *instrument and composition* gate and is unaffected by choosing a global complex-conjugate representation. Likewise, the finite-cycle alternator presupposes a chosen linearization and is compatible with classical completions. Neither changes the mathematical validity of the finite-homogeneity obstruction in its exact-Gromov–Hausdorff scope [2] or the conditional spectrality/strong-symmetry Jordan classifications [3].
The attached exact-symbolic verification covers both counterexamples, all boundary controls, 261 retained rational filter trials (including duplicates and extremely small outliers), and 29 finite cycles, with floating-point residuals recorded separately. The exact proofs, not the grid, establish the universal results. External published sources validate background concepts only; no new experiment, independently refereed proof, instrument calibration, physical time, classical Hamiltonian flow, or total ontology has been provided. Research direction is attributed to Kirtus Justus; internal calculations are AI-assisted provenance, not independent expert corroboration.

References

[1] F. Verstraete, J. Dehaene and B. De Moor, "Local filtering operations on two qubits," *Physical Review A* **64**, 010101(R) (2001). doi:10.1103/PhysRevA.64.010101.

[2] T. Gelander, "Limits of finite homogeneous metric spaces," *L'Enseignement Mathématique* **59**, 195–206 (2013). doi:10.4171/LEM/59-1-8.

[3] H. Barnum and J. Hilgert, "Spectral Properties of Convex Bodies," *Journal of Lie Theory* **30**, 315–344 (2020). doi:10.5802/jolt.1118.